Biology - Undergraduate Level Assessment 8 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. This biome has cold winters and is known for its pine forests.
   (a) Tundra
   (b) Rainforest
   (c) Grassland
   (d) Chaparral
   (e) Savanna
2. Which of these is the smallest unit that natural selection can change?
   (a) an individual's phenotype
   (b) an individual's genotype
   (c) a population's phenotype
   (d) a species' genome
   (e) a species' genotype
3. Why does milk ”when kept in a refrigerator?
   (a) Lactose crystallization at low temperatures causes milk to spoil
   (b) Milk spoils due to absorption of odors in the fridge
   (c) Milk spoils due to the presence of mesophiles
   (d) Milk spoils due to chemical reactions in the fridge
   (e) Thermophiles cause milk to spoil in the refrigerator
4. Over the past two decades retroviruses have profoundly im-pacted the growing field of molecular biology. What retroviral enzyme has most revolutionized this field?
   (a) RNAse H
   (b) Helicase
   (c) integrase
   (d) DNA ligase
   (e) protease
5. Chemical substances released by organisms that elicit a physiological or behavioral response in other members of the same species are known as
   (a) auxins
   (b) neurotransmitters

(c) enzymes
(d) hormones
(e) pheromones

True / False (10 marks)

Answer True or False

6. True or False: Amphibians were the first group of animals to develop lung tissue adequate for supporting their respiratory needs.
7. True or False: The probability that the fourth child will be male is 1/2, as the sex of each child is independent of previous births.
8. True or False: A photosynthetic plant in iron- and magnesium-deficient soil will exhibit dark, large, and robust growth.
9. True or False: Acetylcholine does not function as an intracellular messenger in cellular signaling pathways.
10. True or False: It is commonly believed that around 90% of the human genome encodes proteins.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the concept of heritability using the example of abdominal bristles in Drosophila melanogaster, given that the mean bristles in the parent and F_1 generations were 42.8 and 40.6, respectively. Calculate the heritability of this trait and analyze its implications for understanding genetic inheritance.
12. Discuss the significance of stem reptiles in the evolutionary history of modern reptiles, focusing on their unique lineage and the implications for understanding reptilian evolution.

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